

Jakub Jan Duchniewicz

SENIOR EMBEDDED AND SOFTWARE ENGINEER · TEAM LEAD

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Summary

Engineer working at the intersection of hardware and software, with strong technical, interpersonal, and team leadership skills. My background spans embedded and operating systems, Linux, seL4, FPGAs, low-level performance work, and complex telecommunications systems. Over the past five years, I have focused extensively on 5G and LTE, contributing across modem software, O-RAN systems, ASIC- and FPGA-adjacent development, and hardware/software bring-up.

I am proficient in C, C++, and Rust, with particular interests in hardware acceleration, security, and reverse engineering. As a co-founder of Sticky Piston Studios, I also work at the intersection of technology and creativity. I have presented at conferences including the Embedded Open Source Summit, FOSDEM, and Hackaday Supercon.

Experience

Trustworthy Systems, UNSW

OPERATING SYSTEM ENGINEER

Sydney, Australia

Oct. 2025 – Present

- Developed drivers and board support for the Radxa Rock 3B in seL4 and Microkit-based systems.
- Extended CI coverage and regression testing for the sDDF project.
- Identified and fixed a cross-platform cycle regression caused by incorrect FPU configuration.
- Implemented and integrated an sDDF virtual switch component.

Tietoevry — client engagements: Intel, Sequans, United Micro

SENIOR EMBEDDED AND SOFTWARE ENGINEER

Remote (Wrocław, Poland)

Dec. 2021 – Oct. 2025

- Led, supported, and mentored engineers across embedded and telecommunications projects.
- Implemented 5G features on custom ASIC modem platforms, working across low-level C, firmware, and hardware/software debugging.
- Architected and implemented O-RAN-compliant software, including L2 functionality, FlexRAN-based components, and translation-layer features between IAPI and FAPI.
- Introduced fuzzing of the L1–L2 interface with libFuzzer and investigated issues in DPDK drivers and related networking components.
- **Intel** (Jun. 2022 – Oct. 2023): led the O-RAN M-Plane project, contributed to the FlexRAN 5G solution, implemented L1 and L2 features, performed low-level optimization, and supported hiring and technical leadership.
- **Sequans** (Nov. 2023 – Jun. 2024): designed and delivered a bare-metal modem driver, developed blind decoding for the 5G PDCCH PHY channel, and validated custom ASIC RTL against driver behavior in a real-time system.
- **United Micro** (Jul. 2024 – Jul. 2025): ported firmware and platform code to a 5G eRedCap solution, implemented RSSI scan and L1 synchronization, and debugged RTL and software during bring-up while optimizing memory layout and execution speed.

Sticky Piston Studios

CO-FOUNDER / SOFTWARE ENGINEER

Remote

Dec. 2020 – Present

- Co-founded a studio building products at the intersection of web, game, and embedded technologies.
- Designed and shipped City Event Map, later deployed in the Vilnius region, Lithuania.
- Built full-stack applications spanning frontend, backend, databases, DevOps, and product design.

Google Summer of Code, BeagleBoard.org

EMBEDDED SOFTWARE ENGINEER

Remote

May 2021 – Aug. 2021

- Built a C library for GPGPU acceleration on BeagleBone Black using OpenGL ES 2.0 and EGL.
- Implemented scalar, array, and 2D convolution operations with both single-shot and chained APIs.
- Benchmarked performance across multiple data sizes and documented the project publicly.

Samsung Electronics

Warsaw, Poland / Suwon, South
Korea

JUNIOR SOFTWARE ENGINEER

Dec. 2017 – Mar. 2020

- Improved Tizen broadcast middleware in C++, C, and ARM assembly.
- Optimized performance in a major broadcast pipeline through refactoring and thread-level improvements.
- Built multithreaded middleware and internal Python/Bash tooling for debugging and defect analysis.
- Supported commercialization efforts in South Korea and cross-team defect resolution.

Education

EIT Digital Master School

Europe

M.Sc. IN EMBEDDED SYSTEMS

Sept. 2020 – Nov. 2022

- Recipient of the EIT Excellence Scholarship.
- Winner of the EIT Digital Summer School in Digital Health with *Medpipe*, a personalized recovery tracker for endoprosthetic surgery patients.

KTH Royal Institute of Technology

Stockholm, Sweden

M.Sc. THESIS: FPGA-ACCELERATED PACKET CAPTURE WITH EBPF

Sept. 2021 – Nov. 2022

- Offloaded packet capture from the Linux network stack to FPGA hardware.
- Implemented Avalon-MM host and target interfaces with bursting and stalling support.
- Patched the `stmmac` Ethernet driver and built userspace `pcap` tools for capture and testing.
- Validated the design using ModelSim and SignalTap and achieved performance gains at comparable power.

University of Turku

Turku, Finland

M.Sc. IN EMBEDDED SYSTEMS

Sept. 2020 – Aug. 2021

Warsaw University of Technology

Warsaw, Poland

B.Sc. IN COMPUTER SCIENCE AND NETWORKS

Oct. 2016 – Aug. 2020

- Thesis: FPGA-based hardware accelerator for musical synthesis on Linux.
- Nominated for the IEEE Polish Diploma Contest.
- Implemented DDS synthesis, filtering, polyphony, and waveform generation in Verilog.
- Built Linux kernel and ALSA drivers, plus a userspace MIDI interface for FPGA communication.
- Deployed the solution on the DE0-Nano-SoC platform.

Selected Projects & Leadership

BIBoP

Warsaw, Poland

PROGRAMMING AND ELECTRONICS LEAD

Mar. 2021 – Jun. 2021

- Defined system architecture and sensor data collection for a wearable health prototype.
- Built an ML model for blood pressure prediction from PPG data.
- Deployed the model on AWS Lambda and implemented MQTT-based networking.
- Assembled and soldered the prototype, including analog circuitry.

PolyEngine

Warsaw, Poland

DEVELOPER

Oct. 2017 – Present

- Developed and refactored core game engine architecture, including ECS systems and supporting data structures.
- Built engine and gameplay features during game jams and rapid prototyping projects.
- Contributed to multiple shipped jam games, including Global Game Jam and Slavic Game Jam entries.

KNTG Polygon

Warsaw, Poland

VICE PRESIDENT AND EVENT ORGANISER

Oct. 2017 – Jun. 2019

- Organised weekly meetings for a student game development group.
- Delivered talks on modern C++ and practical language pitfalls.
- Planned Game Dev Fest editions, coordinated invited speakers, and managed sponsorships and promotion.

Selected Presentations

Everything Open

PRESENTER – *Small Game Engines, Big Lessons: PolyEngine and Pill Engine*

[Canberra, ACT, Australia](#)

[Jan. 2026](#)

Hackaday Supercon

PRESENTER – *Hacking an FPGA Design Until It Breaks (and Fixing It Again)*

[Berlin, Germany](#)

[Mar. 2025](#)

FOSDEM

PRESENTER – *Open Source in the Development of 5G and LTE Networks*

[Brussels, Belgium](#)

[Feb. 2024](#)

Embedded Open Source Summit

PRESENTER – *ZLED Frame, or How to Illuminate Your Artwork*

[Seattle, WA, USA](#)

[Apr. 2024](#)

Hackaday Supercon

PRESENTER – *Porting an AI-Powered Wearable Health Monitor to Zephyr on Open Hardware*

[Pasadena, CA, USA](#)

[Nov. 2023](#)

State of Open Con

PRESENTER – *Hardware Acceleration Using FPGAs in Embedded Linux*

[London, United Kingdom](#)

[Mar. 2023](#)

Skills

Languages C, C++, Rust, Python, Bash, ARM/x86/RISC-V assembly, Verilog, SystemVerilog, VHDL

Platforms Embedded systems, Linux kernel, seL4, Microkit, RTOS, bootloaders, microcontrollers, SoCs, ASICs, FPGA

Specialization 5G/LTE L1/L2, O-RAN, network stack, DPDK, hardware acceleration, DSP, GPGPU, fuzzing, reverse engineering

Additional Audio/video processing, game engine development, machine learning, OpenGL ES, CMake, LaTeX

Human Languages Polish, English, German, Spanish

Honors & Awards

Dec. 2023 **Award in Data for City Hackathon 2023**, Urząd m.st. Warszawy

[Warsaw, Poland](#)

Nov. 2022 **Winner – Game Development Challenge**, HackYeah

[Kraków, Poland](#)

2021 **Winner – Best Entrepreneurial Team**, EIT Digital Summer Health School

[Tallinn, Estonia](#)

2019 **Honorable Mention**, Warsaw Film School Game Jam VI

[Warsaw, Poland](#)

2018 **Honorable Mention**, Static Code Analysis Competition – Samsung Electronics

[Global](#)

2017 **2nd Place Overall, 2nd Place in Innovation**, Polyjam / Global Game Jam

[Warsaw, Poland](#)